



Carbon Intelligence & Predictive Eco- Coaching

By John Manoj
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THE CORE CHALLENGE

A Dual Crisis



Climate Degradation

Escalating greenhouse gas emissions.



Public Health Risk

Hazardous co-pollutant exposure.



SOLUTION 01

Health-Linked Calculator

Translating abstract environmental metrics into
personal wellness indicators.

- John Manoj

Bridging Emissions and Health



Carbon Tracking

Tonnes of CO₂e estimated.



Co-Pollutant Mapping

PM_{2.5} and NO_x tracked.



Health Mapping

Personal air quality impact.

Interactive Assessment Modules

Mobility

Evaluating commuting choices.

Energy

Assessing household grid usage.

Diet

Analyzing consumption patterns.



SOLUTION 02

CarbonSense AI Tracker

Leveraging predictive estimation and gamified
behavioural nudges.

BEYOND STATIC NUMBERS

AI-Assisted Nudging

Continuous personalised behavioural modification.

"Transforming occasional awareness into sustained daily action."



Gamified Engagement System



Recommendations

Ranked reduction strategies.



Weekly Streaks

Sustained user participation.



Leaderboards

Institutional benchmark comparisons.

SOLUTION PROPOSAL 01

AI-Driven Personal Eco-Coach

By Joel T Jose

- ✓ **Automated Ingestion:** No manual data entry.
- ✓ **Predictive Engine:** What-if scenario simulator.
- ✓ **ML Personas:** Behavioral pattern recognition algorithms.



Microservices Integration



Decoupled Backend

Node.js and Python FastAPI integration.



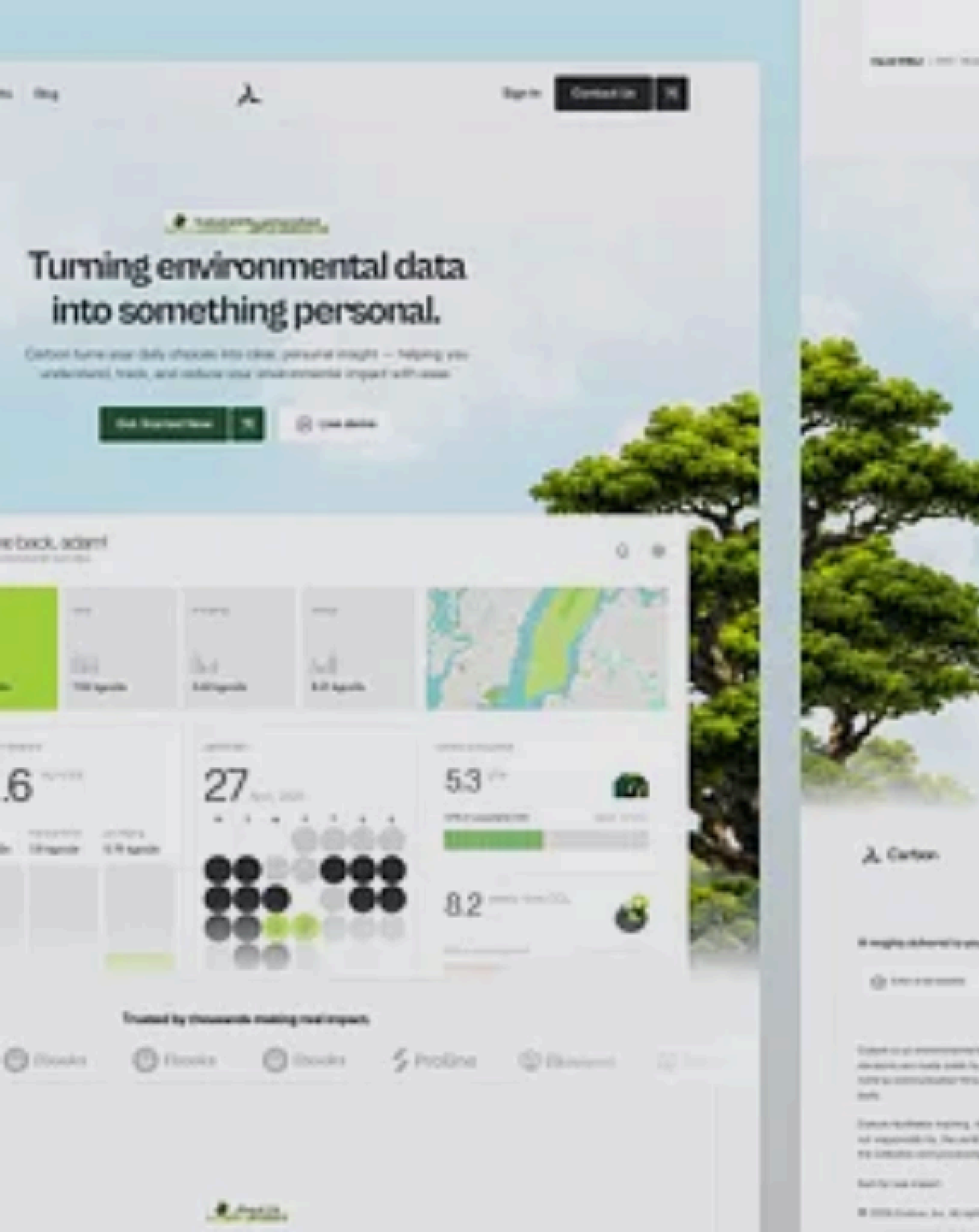
Open LCA Data

Real-time IPCC and DEFRA endpoints.



Predictive ML

K-Means clustering for user personas.



SOLUTION PROPOSAL 02

Web-Based Carbon Calculator

By Johan Byju Jose

5-Category Breakdown

Electricity, transport, food, air travel, waste.

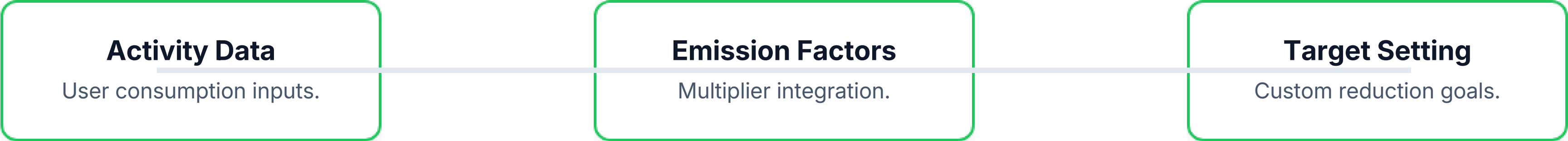
Progress Tracking

Historical comparison of personalized targets.

Accessible Interface

No specialized technical knowledge required.

Calculation Pipeline



SOLUTION PROPOSAL 03

AI-Assisted Nudge Engine

By Joel Sunil

- **Rule-Based Nudges:** Dynamic recommendation engine.
- **Gamification:** Streaks and achievement badges.
- **Offset Market:** Localized carbon-offset integration.





IMPACT SCALING

Behavioral Shifts

Hypothetical Reduction Target

Projected 12% annual reduction per user.

Localized Offsets

Direct funding to local NGO green projects.

Real-Time Feedback

Immediate visual progress indicators.

SOLUTION PROPOSAL 04

Multi-Vector Analytical Engine

By Johann Alphons Cherian

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Core Vectors

ISO

14001 Compliant

EPA

Tree Offsets

Rigorous Compliance



Vector Fusion

Energy, mobility, and consumption unified.



Automated Audits

ISO 14001 continuous alignment protocols.



EPA Equivalencies

Real-world concrete offset translation.

Carbon Tracker

EcoPulse Pro

Designed by Johann Alphons Cherian.

Robustly combining multi-vector localized data, actionable ISO 14001 guidelines, and real-world EPA tree-offset equivalencies into a single interface.



IMPLEMENTATION

EcoPulse Pro Architecture

Dynamic Vectors

Energy, transport, food, shopping, waste calculations.

Impact Badges

Real-time climate hero performance tracking.

Modern UI

Responsive design with dark mode integration.

Simulation & Analytics



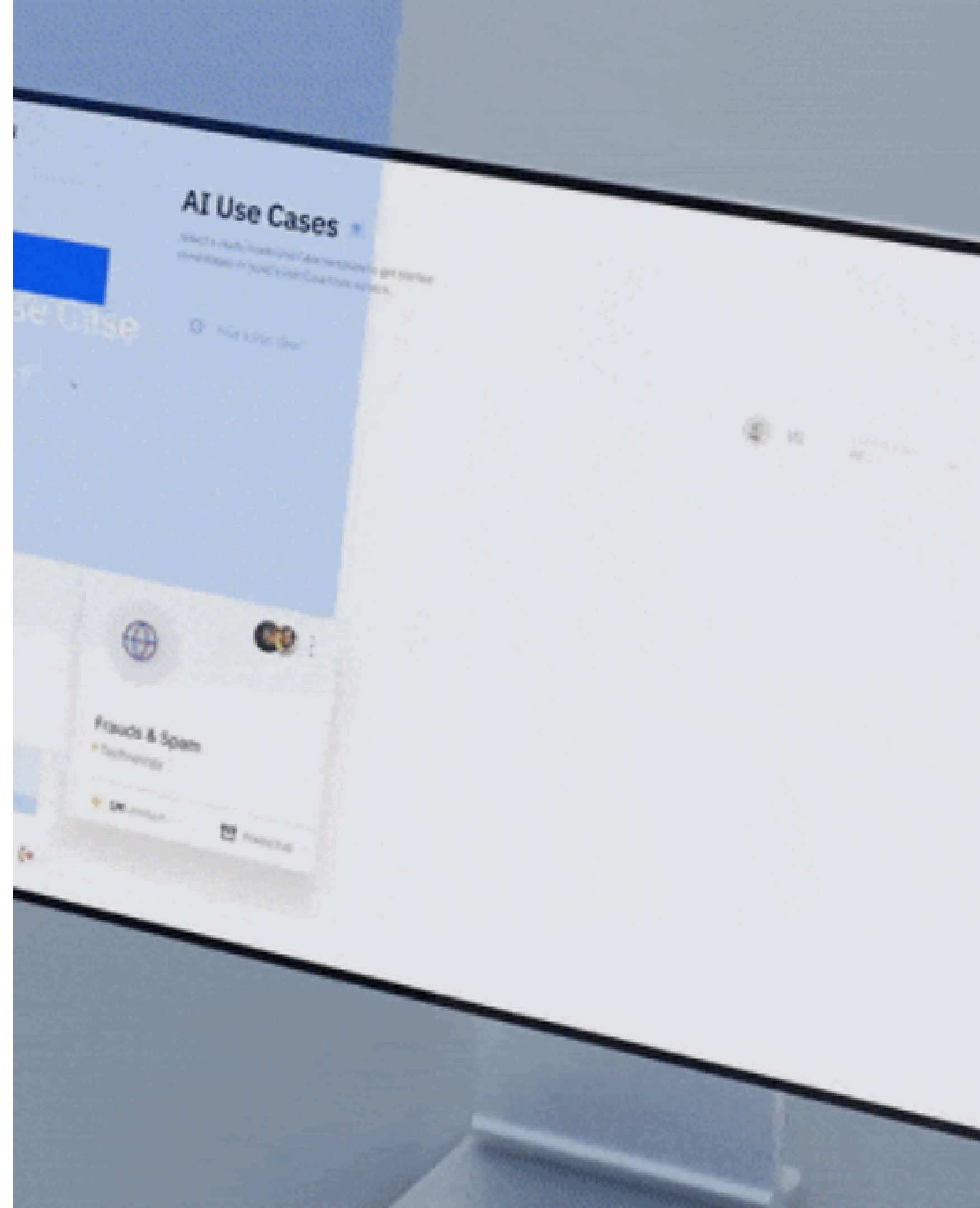
Reduction Pledge Simulator



Real-World Offset Equivalencies



Chart.js KPI Visualizations



Comprehensive Reporting Suite

Emissions Footprint Report

Monthly Scope 1-3 consumption breakdowns.

Offset Verification Report

EPA-backed tree equivalent certificates.

Compliance Audit Report

Automated ISO 14001 alignment documentation.

Predictive Trajectory Report

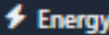
Year-over-year forecasting and trends.



ISO 14001 & EPA FRAMEWORK ALIGNED

Quantify, Compare & Offset Your Environmental Footprint

Comprehensive calculation across home energy, transport, food, shopping, and waste vectors with real-time tree offset equivalencies.



Energy



Transport



Food



Shopping



Waste

REGION GRID FACTOR

Hydro/Nuclear (e.g., Norway/Iceland) - 0.0



HOUSEHOLD SIZE

13

Home Utilities & Fuels

Electricity (kWh/month)

442

Natural Gas (\$ or Therms/month)

40

LPG Gas (Cylinders 14.2kg/mo)

20

Heating Oil (Liters/month)

20



Rooftop Solar Installed (Offsets 60% Electricity Carbon)

REDUCTION GOAL SIMULATOR

30% Reduction Target

Simulate actionable lifestyle reductions to see immediate CO₂ and tree savings.

Reset Defaults

Auto-saved to browser storage

ANNUAL INDIVIDUAL EMISSION DENSITY

3.32 tCO₂e / year

Moderate Impact (Global Avg)

Paris Target (2.0t)

100%



REAL-WORLD OFFSET EQUIVALENCIES



152

Trees Needed to Offset / Yr



8,210

Gas Miles Driven



403.5

Phones Charged (k)



Category Emission Breakdown



Energy Transport Food
Shopping Waste

Solution of the Problem

Live Demonstration



Transitioning to the live evaluation of the EcoPulse Pro portal architecture and analytics suite.

Thank You

Image Sources



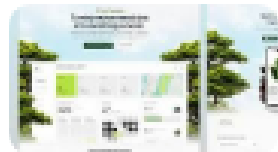
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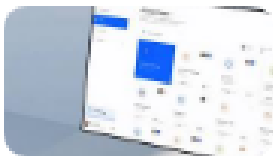
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Image Sources



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